

Rajalakshmi Engineering College

Name: Mahathi B

Email: mahathi.b.2024.csd@rajalakshmi.edu.in

Roll no: 241701029

Phone: 9841411040

Branch: REC

Department: CSD

Batch: 2028

Degree: B.E - CSD

Scan to verify results



2024_28_IV_CSD_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 2_CY

Attempt : 1

Total Mark : 40

Marks Obtained : 36

Section 1 : Coding

1. Problem Statement

Joe has a favourite number, let's call it X. He wants to check if X is divisible by the sum of its digits. If it is, he considers it a lucky number. If not, he wants to find the closest smaller number, that is divisible by the sum of digits of X. Joe has challenged his friends to solve this puzzle at his birthday party.

Example

Input:

157

Output:

157 is not divisible by the sum of its digits.

The closest smaller number that is divisible: 156

Explanation:

The sum of the digits of X is $1+5+7=13$. Since 157 is not divisible by 13, we need to find the closest smaller number that is divisible by 13. 156 is divisible by 13, it is the closest smaller number that meets the requirement.

Input Format

The input consists of an integer X, representing Joe's favourite number.

Output Format

If X is a lucky number, then the output must be in the format: "X is divisible by the sum of its digits."

If not, then the output must be in the format:

"X is not divisible by the sum of its digits."

The closest smaller number that is divisible: Y",

where X is the entered number and Y is the closest number.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 120

Output: 120 is divisible by the sum of its digits.

Answer

```
// You are using Java
import java.util.Scanner;
class lucky{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int temp=n;
        int sum=0;
        while(temp>0){
```

```

        sum+=temp%10;
        temp=temp/10;
    }
    if(n%sum==0){
        System.out.printf("%d is divisible by the sum of its digits.",n);
    }
    else{
        System.out.printf("%d is not divisible by the sum of its digits.\n",n);
        for(int i=n-1 ; i>=1 ; i--){
            int temp1=i;
            int sum1 =0;
            while(temp1>0){
                sum1+=temp1%10;
                temp1/=10;
            }
            if(i%sum1==0 && i%sum1==0){
                System.out.printf("The closest smaller number that is divisible: %d",i);
                break;
            }
        }
    }
}
}
}

```

Status : Partially correct

Marks : 6/10

2. Problem Statement

Samantha is a diligent math student who is exploring the world of programming. She is learning Java and has recently studied conditional statements. One day, her teacher gives her an interesting problem to solve, which takes a number as input and checks whether it is a multiple of 5 or 7.

Help her complete the task.

Input Format

The input consists of a single integer N, representing the number to be checked.

Output Format

If the number is a multiple of 5 but not 7, the output prints "N is a multiple of 5".

If the number is a multiple of 7, the output prints "N is a multiple of 7".

Otherwise the output prints "N is neither multiple of 5 nor 7" where N is an entered integer.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 10

Output: 10 is a multiple of 5

Answer

// You are using Java

```
import java.util.Scanner;
```

```
class mul{
```

```
    public static void main(String[] args){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
```

```
        if(n%5==0 && n%7!=0){
```

```
            System.out.printf("%d is a multiple of 5",n);
```

```
        }
```

```
        else if(n%7==0 && n%5!=0){
```

```
            System.out.printf("%d is a multiple of 7",n);
```

```
        }
```

```
        else{
```

```
            System.out.printf("%d is neither multiple of 5 nor 7",n);
```

```
        }
```

```
    }
```

```
}
```

Status : Correct

Marks : 10/10

3. Problem Statement

Noah is analyzing numbers within a given range [A, B] and wants to calculate a special sum. For each number in the range, he calculates the product of its odd digits (ignoring even digits). If the number contains no odd digits, it is skipped. The sum of these products for all numbers in the range is the result.

Write a program to compute this sum.

Example

Input:

10 12

Output:

3

Explanation:

For 10, odd digits = 1, product = 1.

For 11, odd digits = 1, 1, product = $1 * 1 = 1$.

For 12, odd digits = 1, product = 1.

Total sum = $1 + 1 + 1 = 3$

Input Format

The input consists of two space-separated integers A and B, representing the inclusive range boundaries.

Output Format

The output prints a single integer representing the sum of the products of odd digits for all numbers in the range.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 10 12

Output: 3

Answer

```
import java.util.Scanner;
class sum{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        int sum=0;
        for(int i=a ; i<=b ; i++){
            int temp=i;
            int pro1=0;

            while(temp>0){
                int x=temp%10;
                if(x%2!=0){
                    if(pro1==0){
                        pro1=x;
                    }
                    else{
                        pro1*=x;
                    }
                }
                temp/=10;
            }
            sum+=pro1;
        }
        System.out.printf("%d",sum);
    }
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Ted, the computer science enthusiast, has accepted the challenge of writing a program that checks if the number of digits in an integer matches the sum of its digits.

Guide Ted in designing and writing the code to solve this problem using a 'do-while' loop.

Input Format

The input consists of an integer N, representing the number to be checked.

Output Format

If the sum is equal to the number of digits, print "The number of digits in N matches the sum of its digits."

Else, print "The number of digits in N does not match the sum of its digits."

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 20

Output: The number of digits in 20 matches the sum of its digits.

Answer

```
// You are using Java
import java.util.Scanner;
class sum{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int temp=a;
        int sum=0;
        int c =0;
        while(temp>0){
            sum+=temp%10;
            temp/=10;
            c++;
        }
        if(sum==c){
            System.out.printf("The number of digits in %d matches the sum of its
digits.",a);
        }
        else{
```

```
System.out.printf("The number of digits in %d does not match the sum of  
its digits.",a);  
}
```

```
}  
}
```

Status : Correct

Marks : 10/10